

Solución

- a) La distribución conjunta: $f(x_1, \dots, x_n | (\mu, \sigma^2)) = \prod_{i=1}^n f(x_i | (\mu, \sigma^2))$
 donde $f(x_i) \sim N(\mu, \sigma^2)$ por ende } 10 pts

La familia de Distribuciones conjuntas viene dada. Por

$$\left\{ \left(\frac{1}{\sqrt{2\pi\sigma^2}} \right)^n \cdot e^{-\frac{\sum (x_i - \mu)^2}{2\sigma^2}}, \mu \in \mathbb{R} \text{ y } \sigma^2 > 0 \right\} \quad \left. \vphantom{\left(\frac{1}{\sqrt{2\pi\sigma^2}} \right)^n} \right\} 20 \text{ pts}$$

- b) Note que $E(\hat{\theta}) = E(\alpha \bar{X}^2) = \alpha \cdot E(\bar{X}^2)$ } 5 pts

Considere que $\text{Var}(\bar{X}) = E(\bar{X}^2) - E(\bar{X})^2$

$$\Rightarrow \frac{\sigma^2}{n} = E(\bar{X}^2) - \mu^2$$

$$\Rightarrow \frac{\sigma^2}{n} + \mu^2 = E(\bar{X}^2)$$

} 20 pts

$$\Rightarrow E(\alpha \bar{X}^2) = \alpha \cdot \left(\frac{\sigma^2}{n} + \mu^2 \right) = \frac{\alpha \sigma^2}{n} + \alpha \mu^2$$

$\therefore \hat{\theta} = \alpha \bar{X}^2$ es sesgado para $\theta = \alpha \mu^2$ y su sesgo es respectivamente $\frac{\alpha \cdot \sigma^2}{n}$ } 5 pts

Opción 1

c) $\lim_{n \rightarrow \infty} E(\hat{\theta}) \stackrel{?}{=} \theta$ (Hay que verificar esto)

$$\Rightarrow \lim_{n \rightarrow \infty} \frac{\alpha \cdot \sigma^2}{n} = 0$$

} 20 pts

Opción 2.

$$\lim_{n \rightarrow \infty} E(\hat{\theta}) = \theta$$

$$\Rightarrow \lim_{n \rightarrow \infty} \frac{\alpha \sigma^2}{n} + \alpha \mu^2 = \alpha \mu^2 = \theta$$

} 20 pts.

$\therefore \hat{\theta}$ es asintóticamente insesgado. } 5 pts $\therefore \theta$ es asintóticamente insesgado

d)

$$\text{Sea } \hat{\theta}_1 = \frac{\hat{\theta}}{\alpha} - \frac{6^2}{n}$$

$$\text{Observe que } E(\hat{\theta}_1) = E\left(\frac{\hat{\theta}}{\alpha} - \frac{6^2}{n}\right) = \frac{1}{\alpha} \cdot E(\hat{\theta}) - \frac{6^2}{n}$$

$$= \frac{1}{\alpha} (\alpha \cdot \mu^2 + \alpha \frac{6^2}{n}) - \frac{6^2}{n} = \mu^2 //$$

20 pts